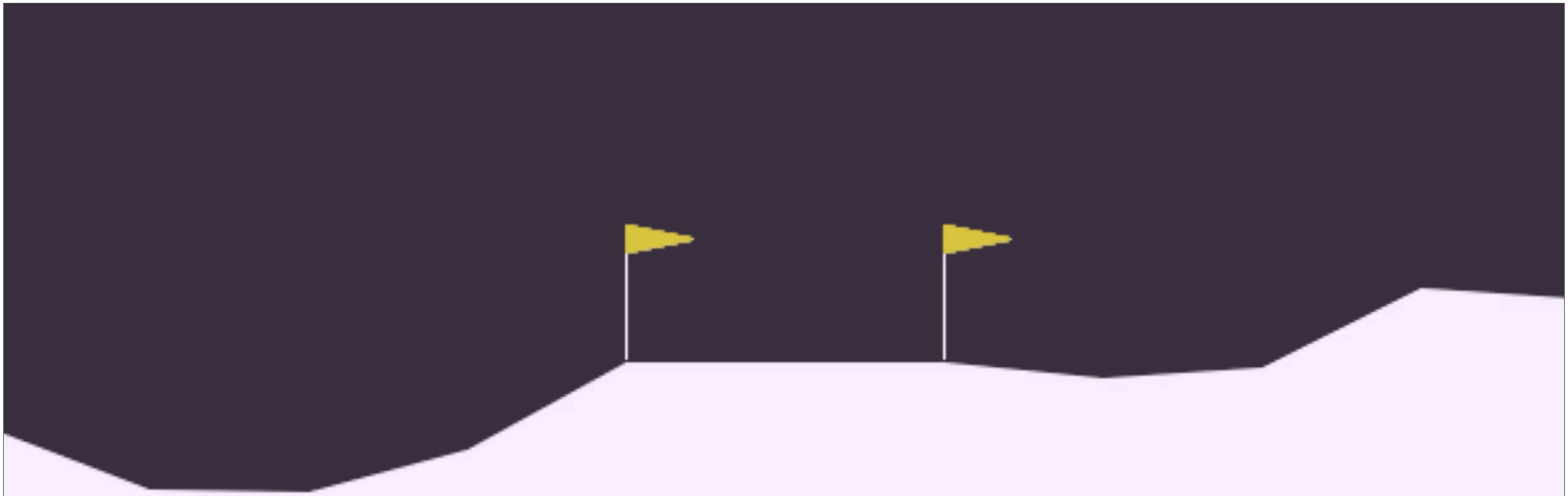
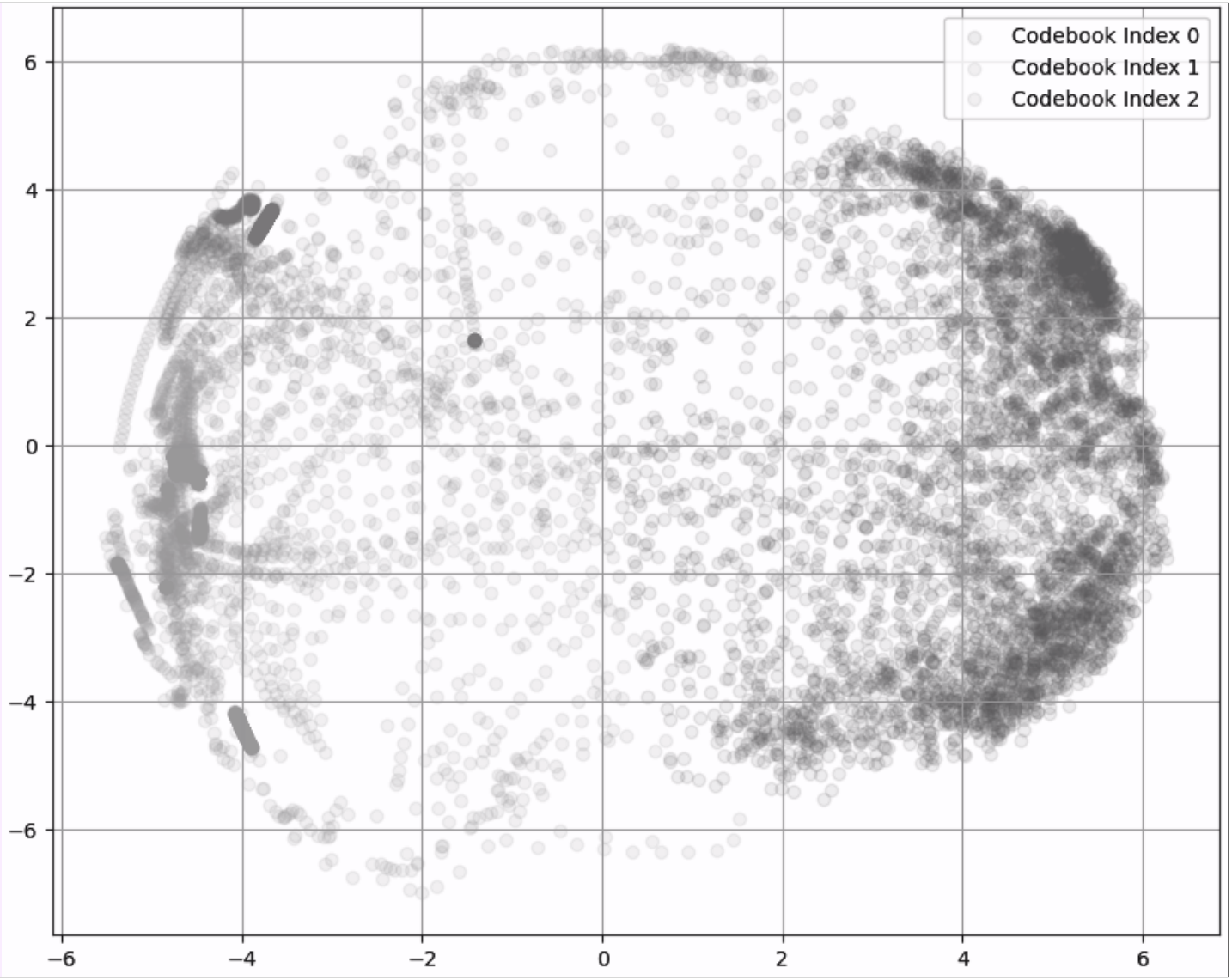


Interpretable Reinforcement Learning through Clustering-Enhanced Layer Regularisation

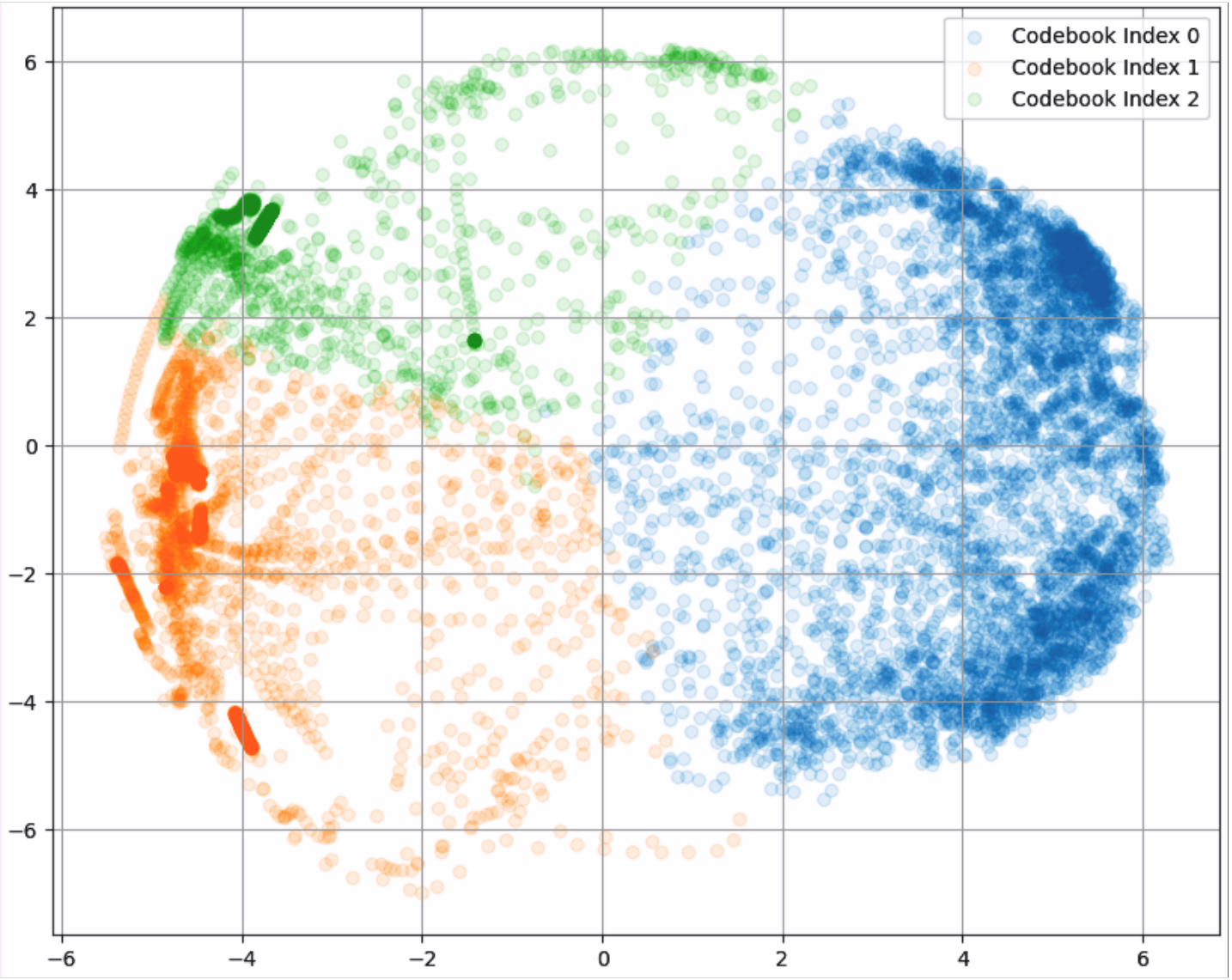
Leonardo Barazza, 10/09/2024

What Do Hidden Layers Look Like?



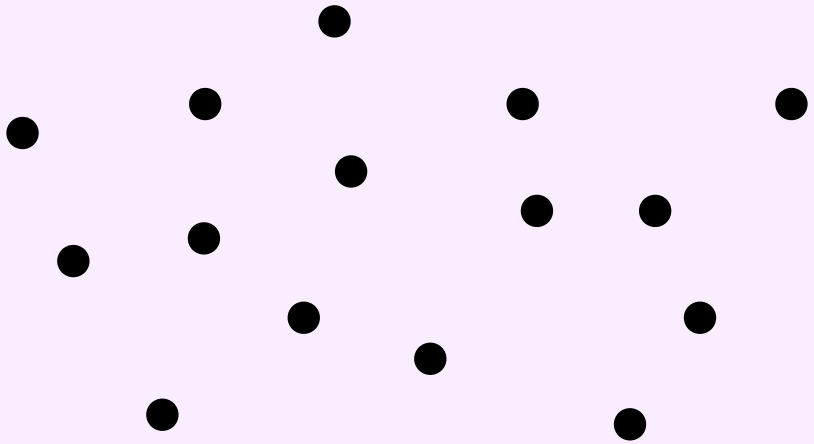
PCA hidden layer visualization of a baseline agent trained on LunarLander.

What Do Hidden Layers Look Like?



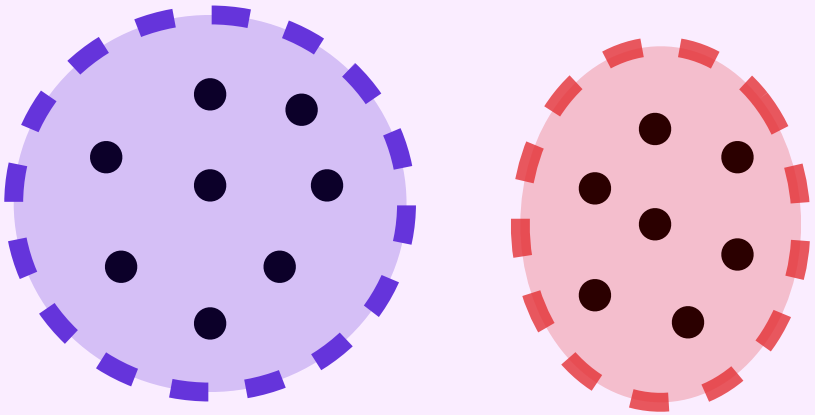
PCA hidden layer visualization of a baseline agent trained on LunarLander.

What If We Could Induce States to Cluster Together?



Hidden Layer

What If We Could Induce States to Cluster Together?



Hidden Layer

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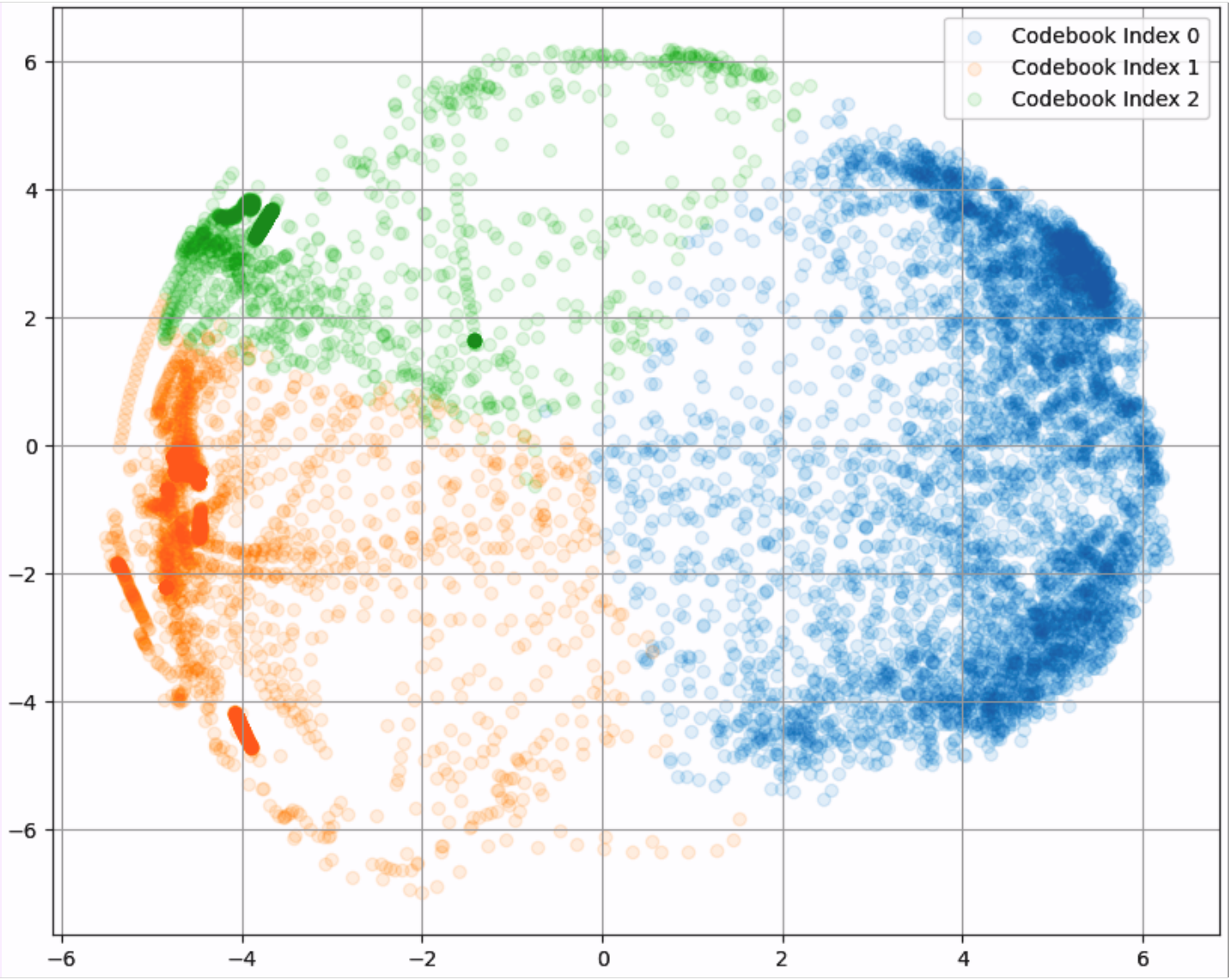
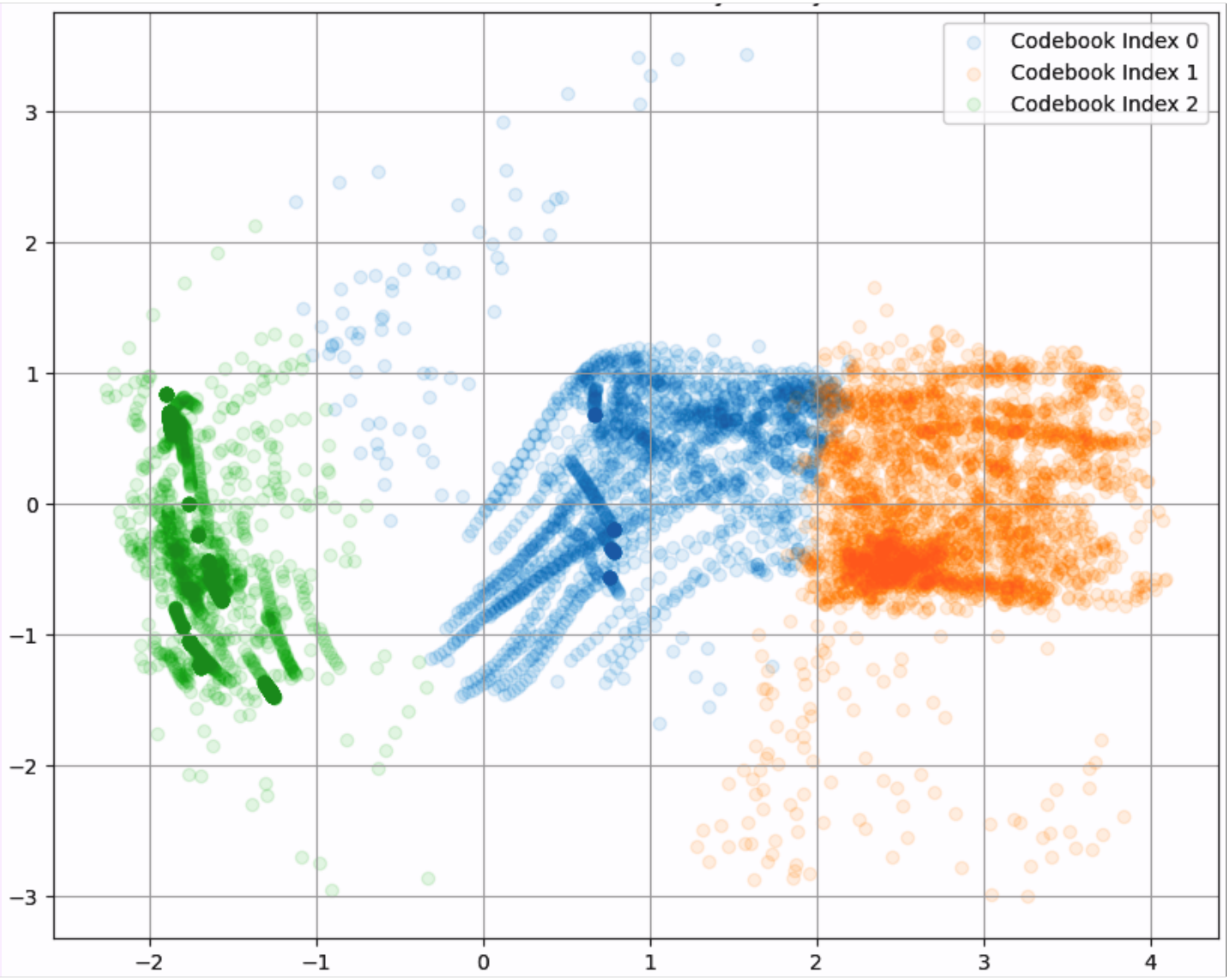


PCA hidden layer visualization of a baseline agent trained on LunarLander.

What If We Could Induce States to Cluster Together?



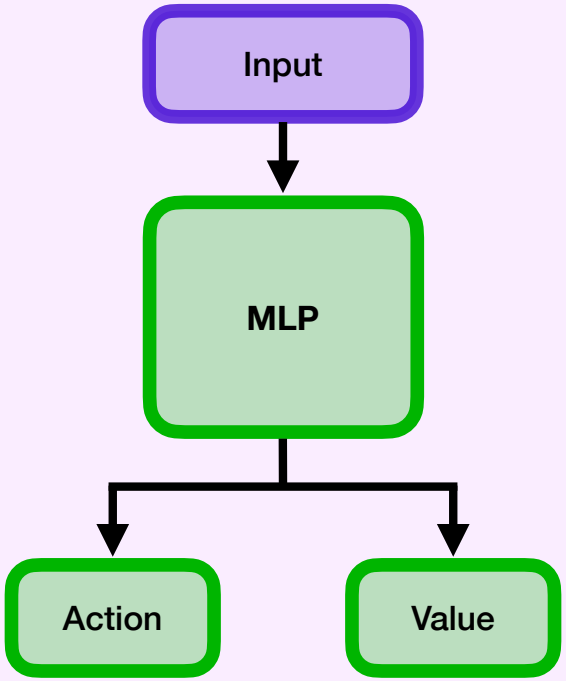
PCA hidden layer visualization of a CELaRe agent trained on LunarLander.

Contributions

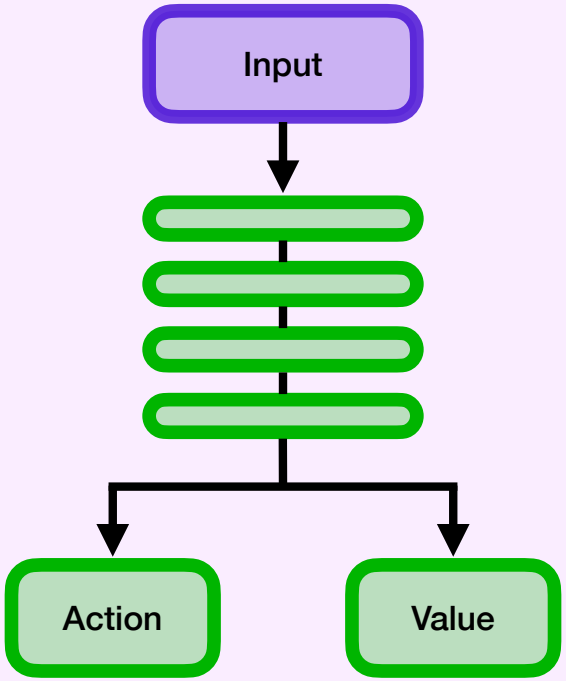
1. The CELaRe Architecture
2. A Novel Objective
3. Quantifiable Metrics
4. The Oscillating Coefficient
5. Reading the Hidden Layer
6. Experimental Results

The CELaRe Architecture

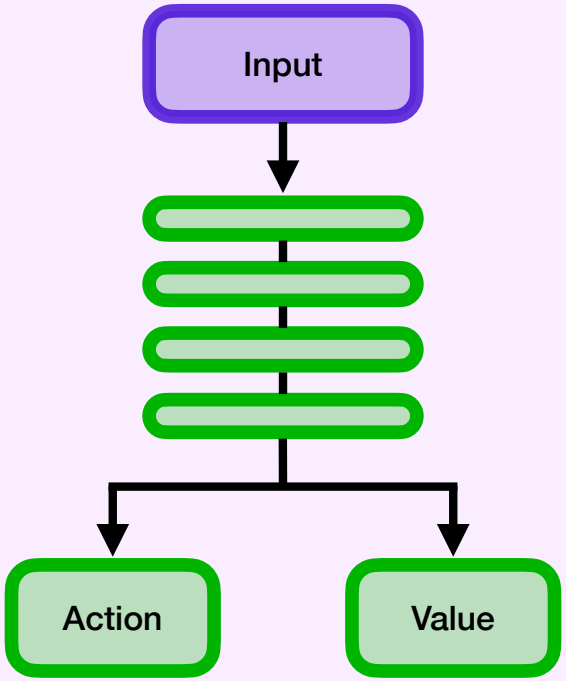
The Baseline Architecture



The Baseline Architecture



The Baseline Architecture



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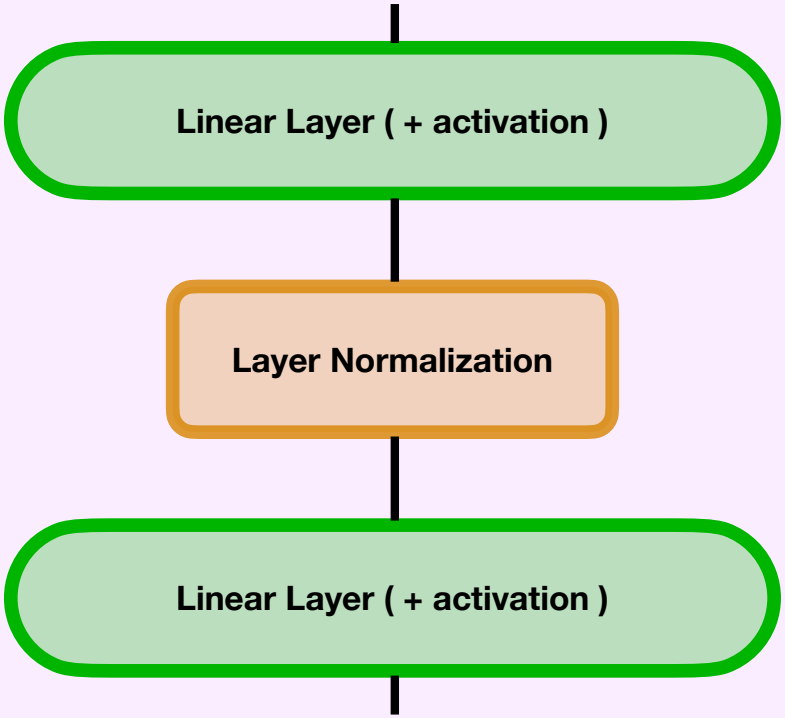
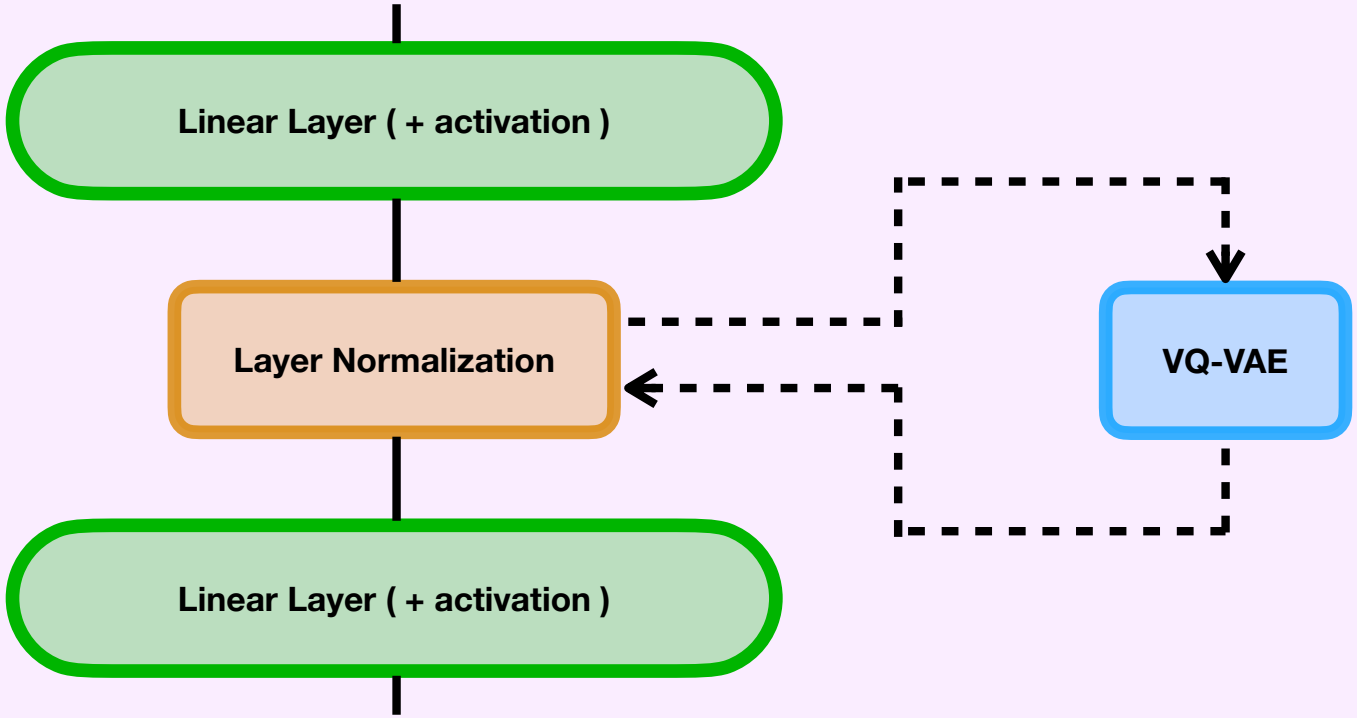


The CELaRe Architecture



Contributions

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A Novel Objective

$$\mathcal{L}_{\text{cluster}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{quantization}} - \alpha \cdot \mathcal{L}_{\text{entropy}}$$

A Novel Objective

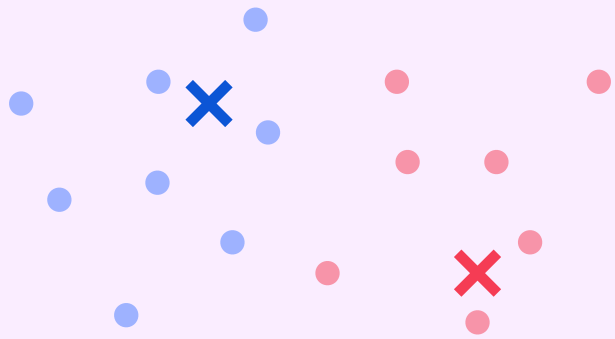
Reconstruction Loss

$$\mathcal{L}_{\text{reconstruction}} = \|\text{s.g.}(x) - D(\boldsymbol{e}_k)\|^2 + \|x - \text{s.g.}(D(\boldsymbol{e}_k))\|^2$$

A Novel Objective

Reconstruction Loss

Hidden Layer

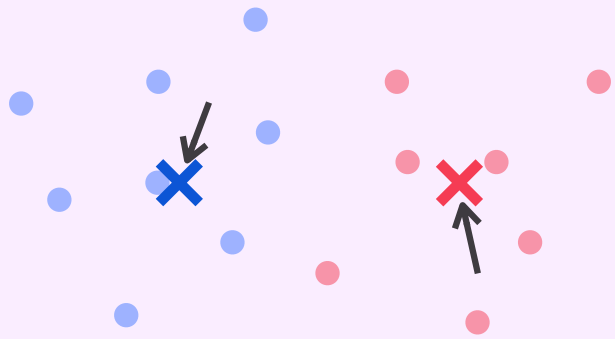


$$\mathcal{L}_{\text{reconstruction}} = \|\text{s.g.}(x) - D(\boldsymbol{e}_k)\|^2 + \|x - \text{s.g.}(D(\boldsymbol{e}_k))\|^2$$

A Novel Objective

Reconstruction Loss

Hidden Layer

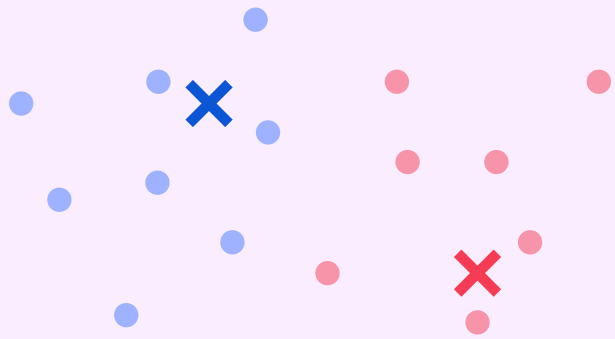


$$\mathcal{L}_{\text{reconstruction}} = \|\text{s.g.}(x) - D(\boldsymbol{e}_k)\|^2 + \|x - \text{s.g.}(D(\boldsymbol{e}_k))\|^2$$

A Novel Objective

Reconstruction Loss

Hidden Layer

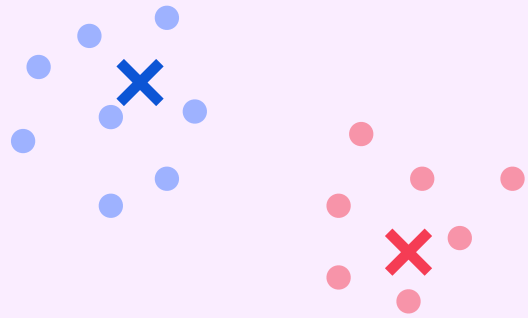


$$\mathcal{L}_{\text{reconstruction}} = \|\text{s.g.}(x) - D(\boldsymbol{e}_k)\|^2 + \|x - \text{s.g.}(D(\boldsymbol{e}_k))\|^2$$

A Novel Objective

Reconstruction Loss

Hidden Layer



$$\mathcal{L}_{\text{reconstruction}} = \|\text{s.g.}(x) - D(\boldsymbol{e}_k)\|^2 + \|x - \text{s.g.}(D(\boldsymbol{e}_k))\|^2$$

A Novel Objective

$$\mathcal{L}_{\text{cluster}} = \mathcal{L}_{\text{reconstruction}} + \mathcal{L}_{\text{quantization}} - \alpha \cdot \mathcal{L}_{\text{entropy}}$$

A Novel Objective

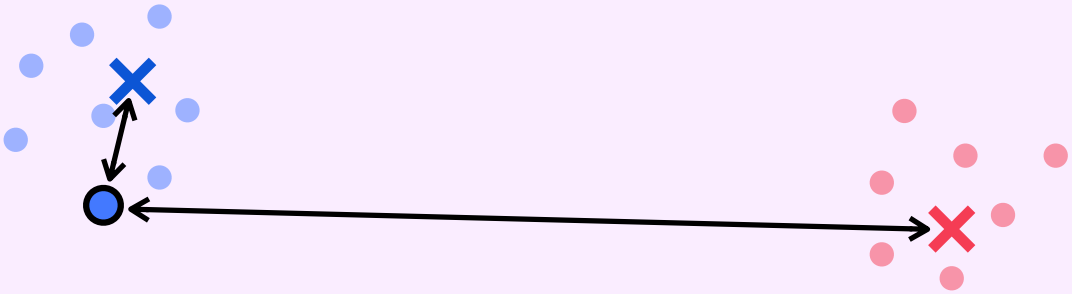
Entropy Loss

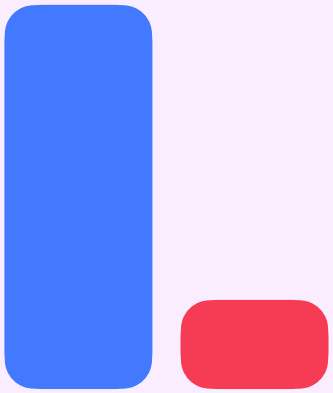
$$\mathcal{L}_{\text{entropy}} = -\sum_{j} u_j \log(u_j + \epsilon)$$

A Novel Objective

Entropy Loss

Latent Space





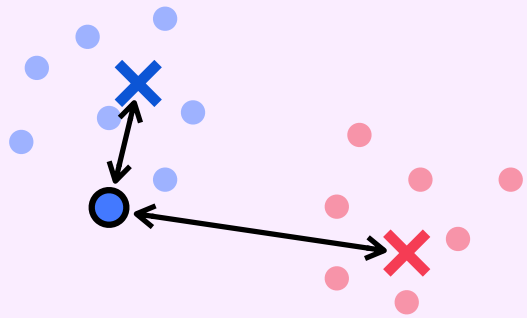
Probabilities

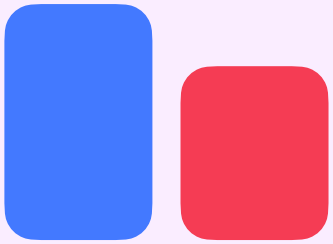
$$\mathcal{L}_{\text{entropy}} = -\sum_j u_j \log(u_j + \epsilon)$$

A Novel Objective

Entropy Loss

Latent Space





Probabilities

$$\mathcal{L}_{\text{entropy}} = -\sum_j u_j \log(u_j + \epsilon)$$

Contributions

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2. A Novel Objective
3. Quantifiable Metrics
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6. Experimental Results

Quantifiable Metrics

1. Performance
2. Intepretability

Quantifiable Metrics

1. Performance $\longrightarrow$ Episodic Return
2. Intepretability

Quantifiable Metrics

1. Performance → Episodic Return
2. Intepretability → Quantization Loss

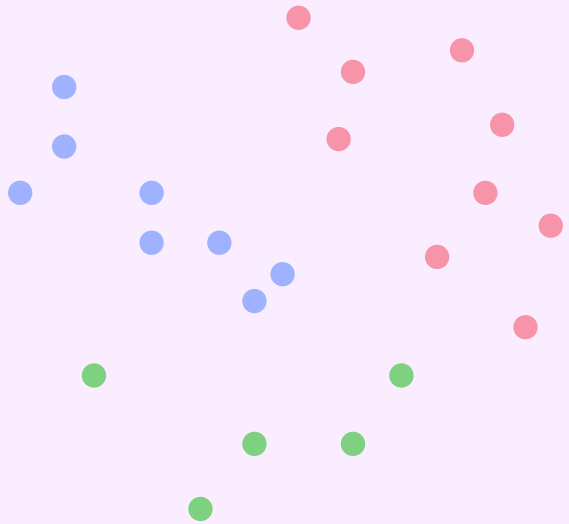
Quantifiable Metrics

Quantization Loss

$$\mathcal{L}_{\text{quantization}} = \|\text{s.g.}(E(x)) - \boldsymbol{e}_k\|^2 + \beta \cdot \|E(x) - \text{s.g.}(\boldsymbol{e}_k)\|^2$$

Quantifiable Metrics

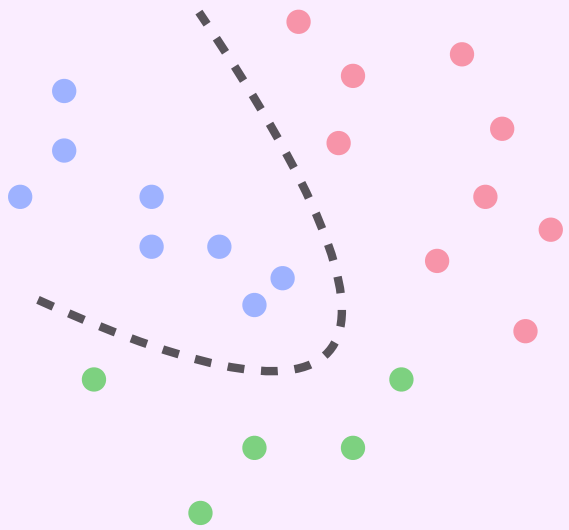
Quantization Loss



Hidden Layer

Quantifiable Metrics

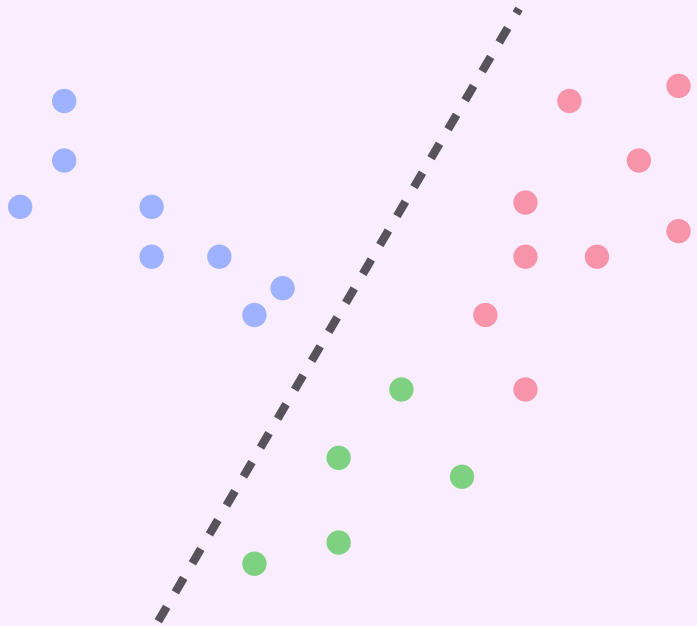
Quantization Loss



Hidden Layer

Quantifiable Metrics

Quantization Loss



Hidden Layer

Contributions

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The Oscillating Coefficient

The Oscillating Coefficient

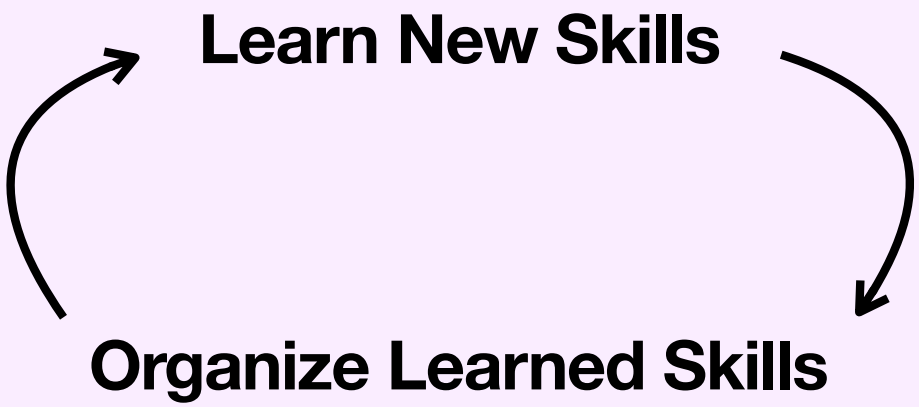
Learn New Skills

The Oscillating Coefficient

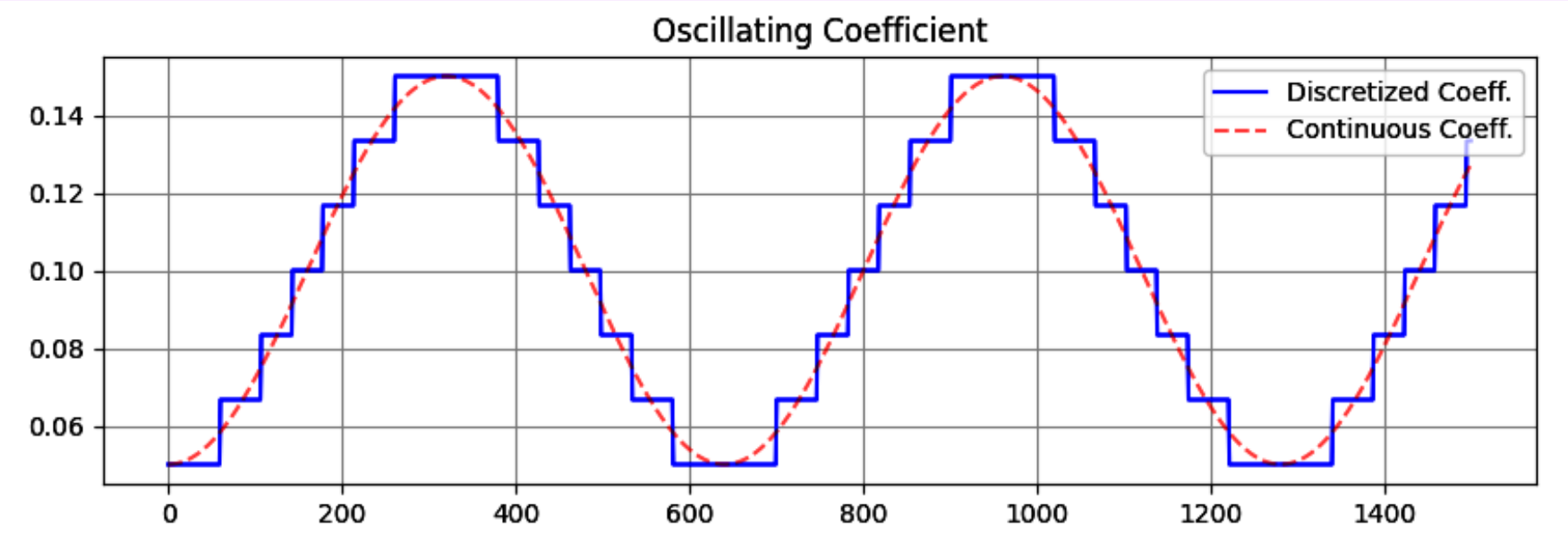
Learn New Skills

Organize Learned Skills

The Oscillating Coefficient



The Oscillating Coefficient



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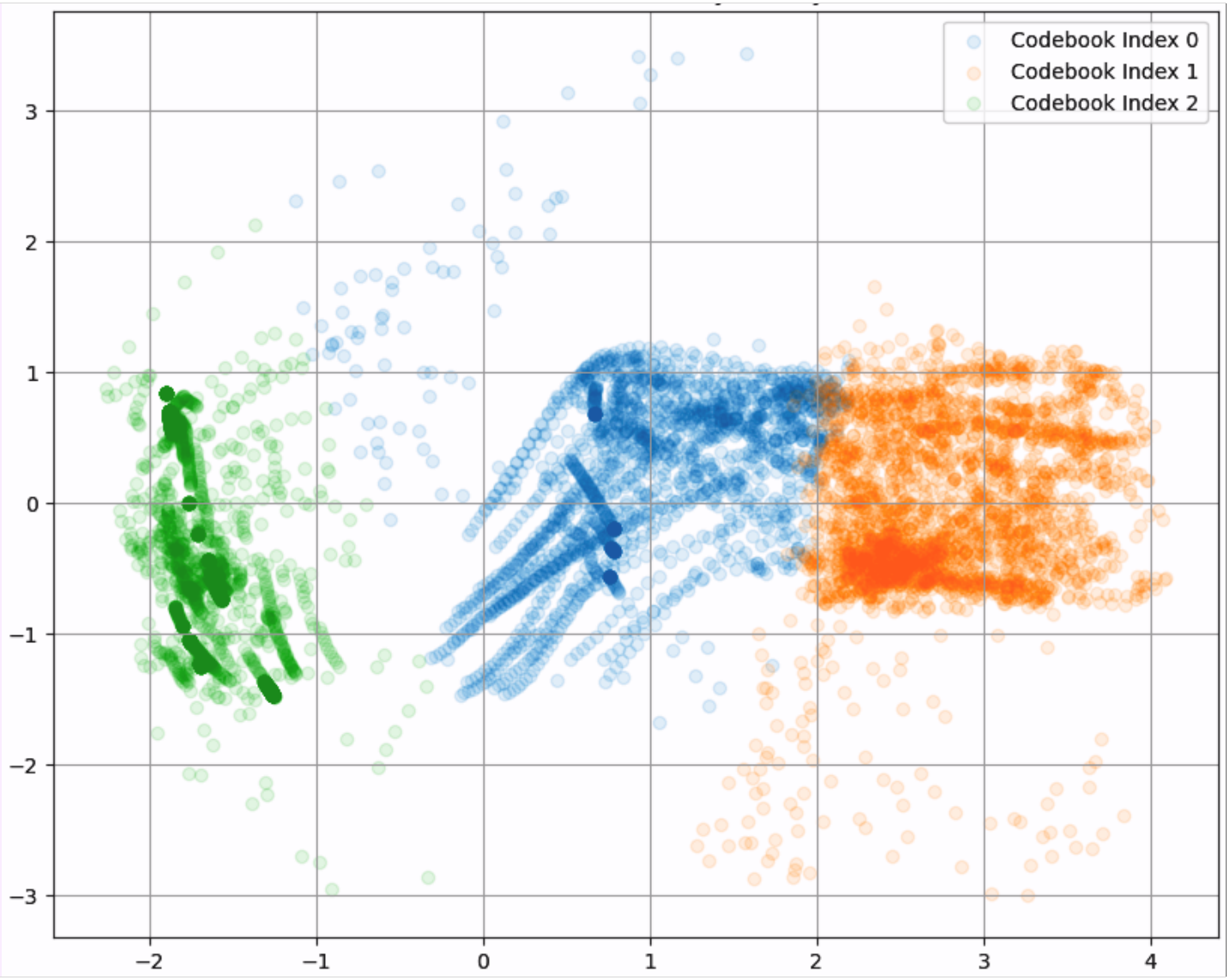
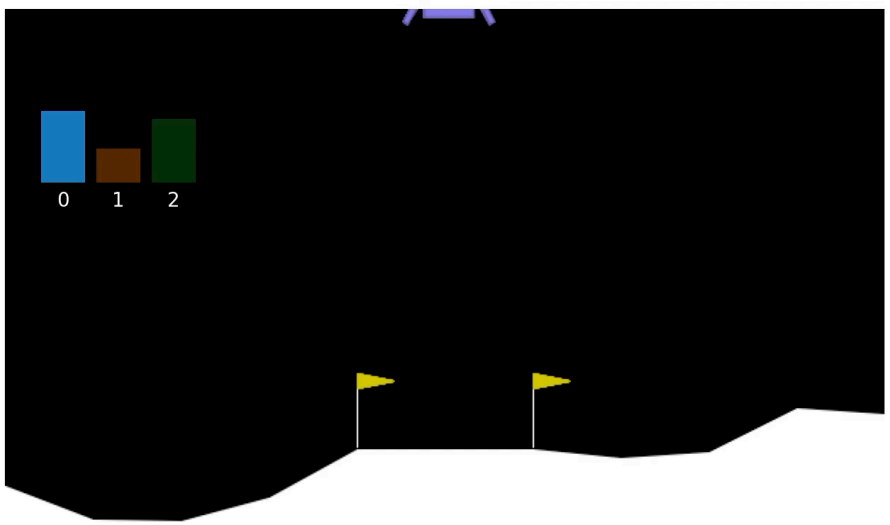
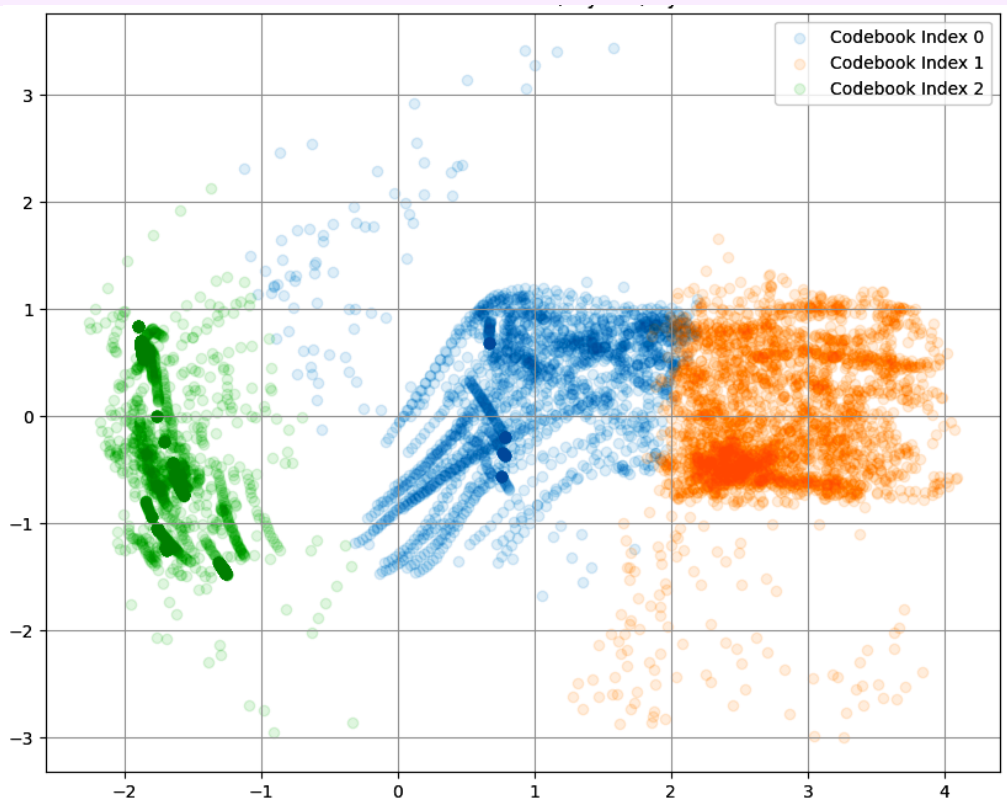
Reading the Hidden Layer



PCA hidden layer visualization of a CELaRe agent trained on LunarLander.

Reading the Hidden Layer





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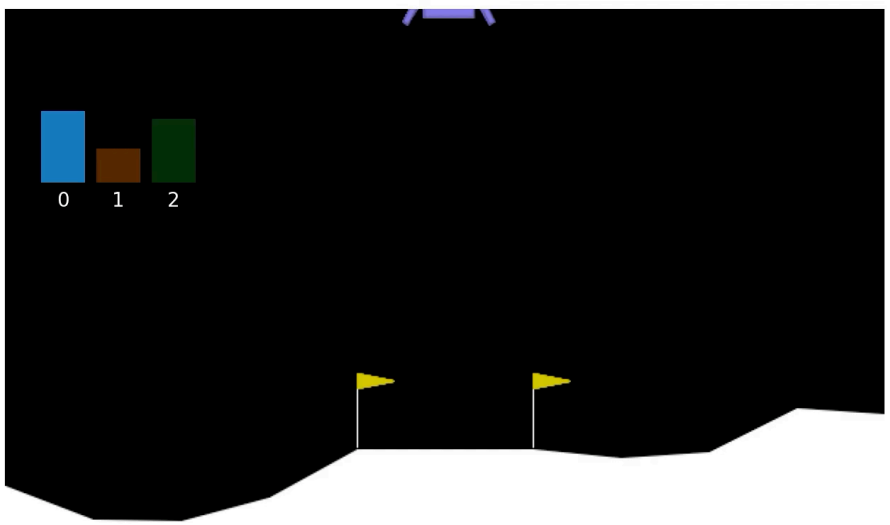

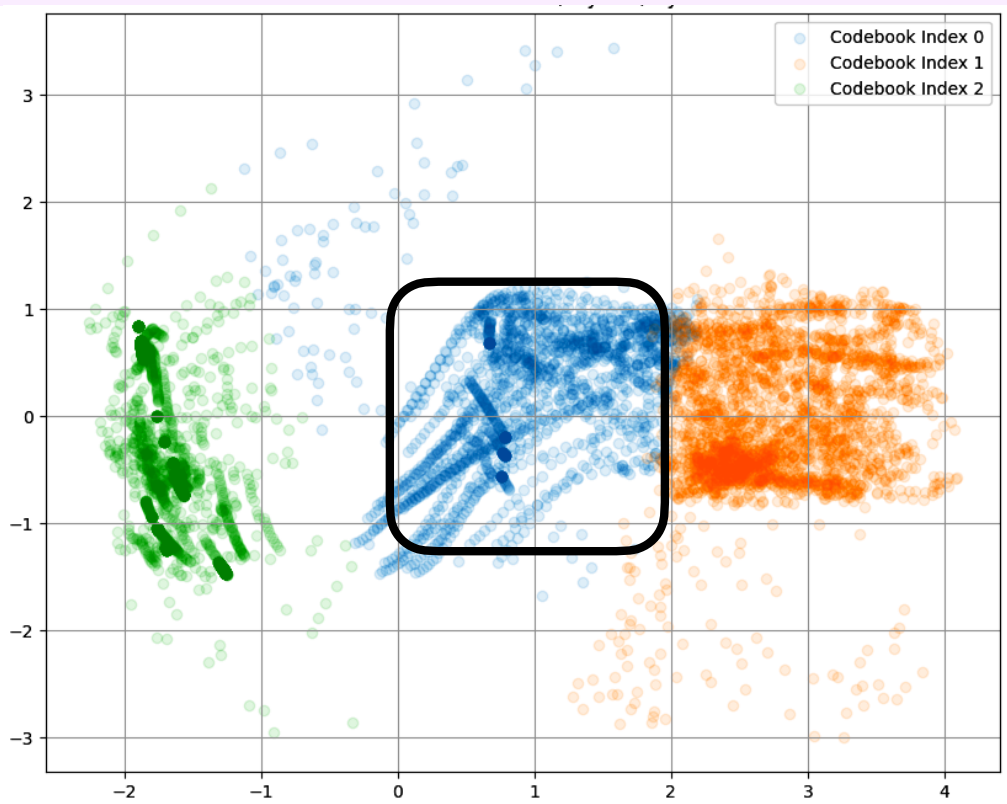

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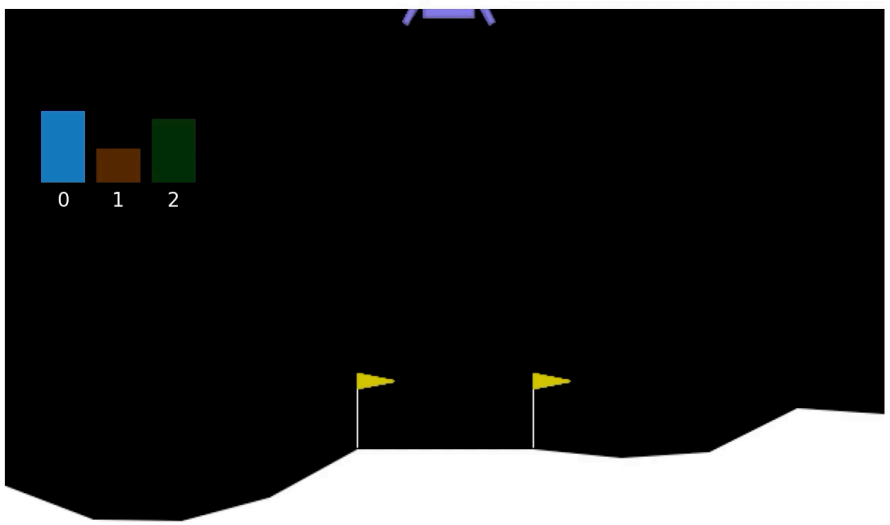

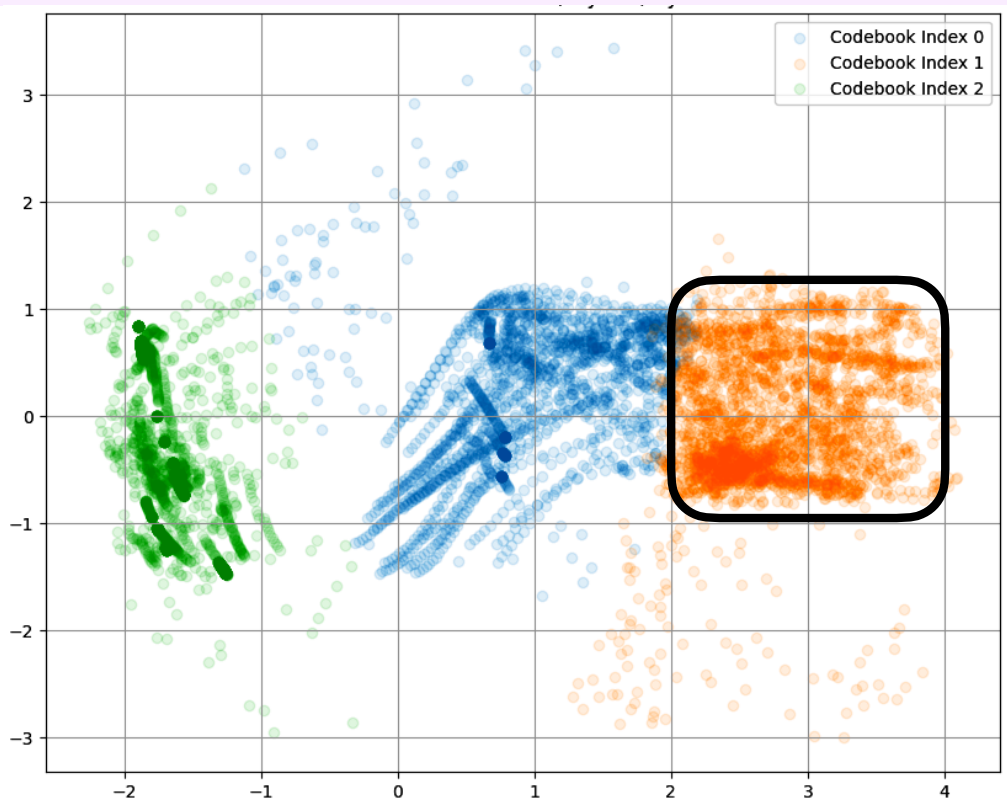

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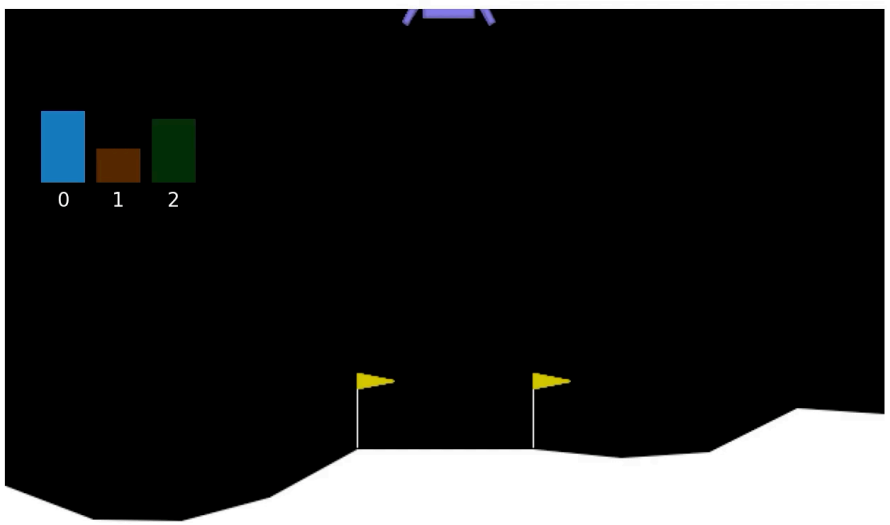

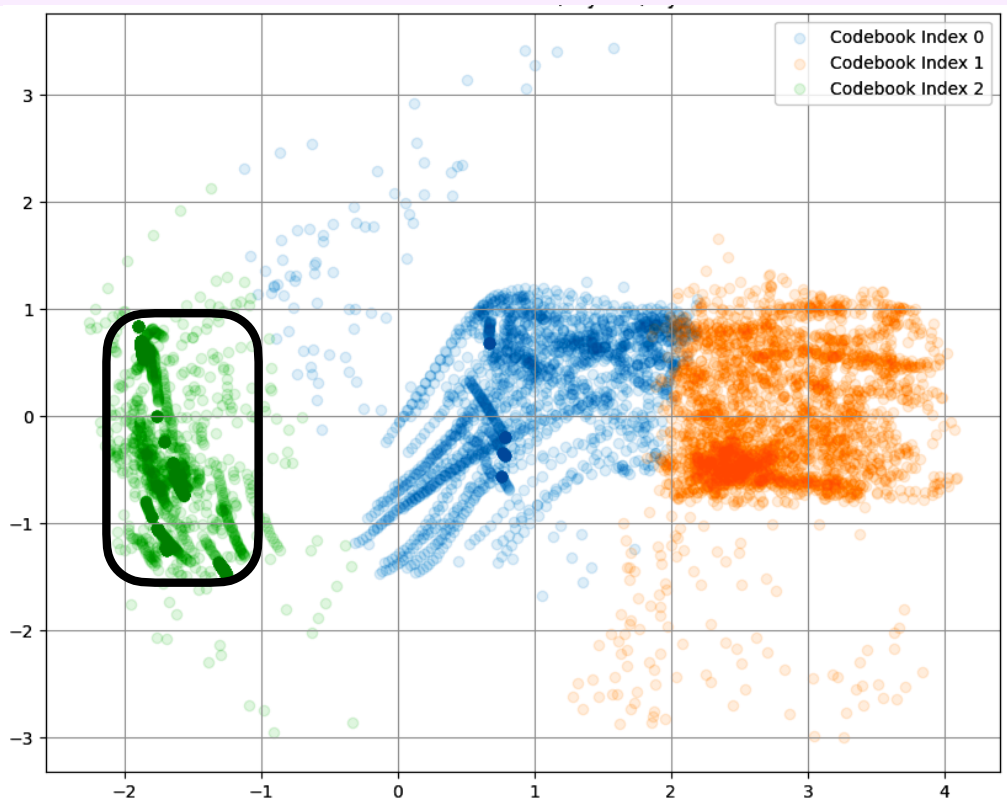

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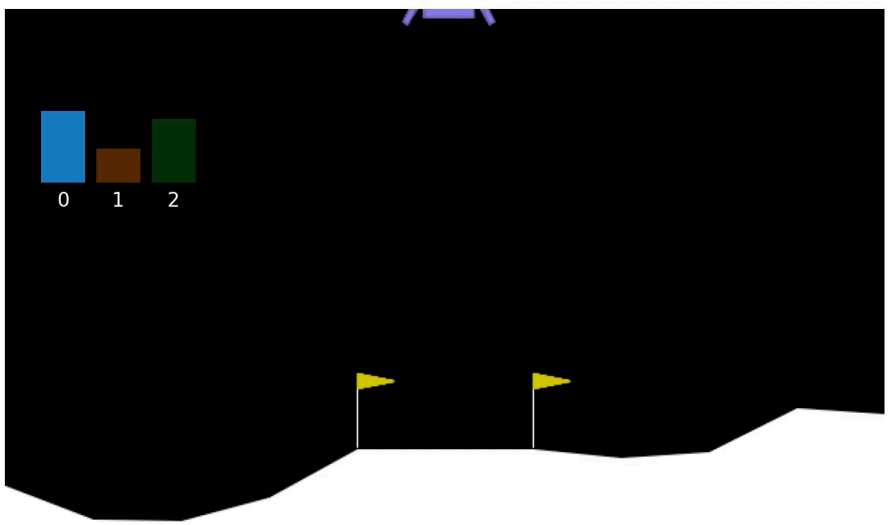

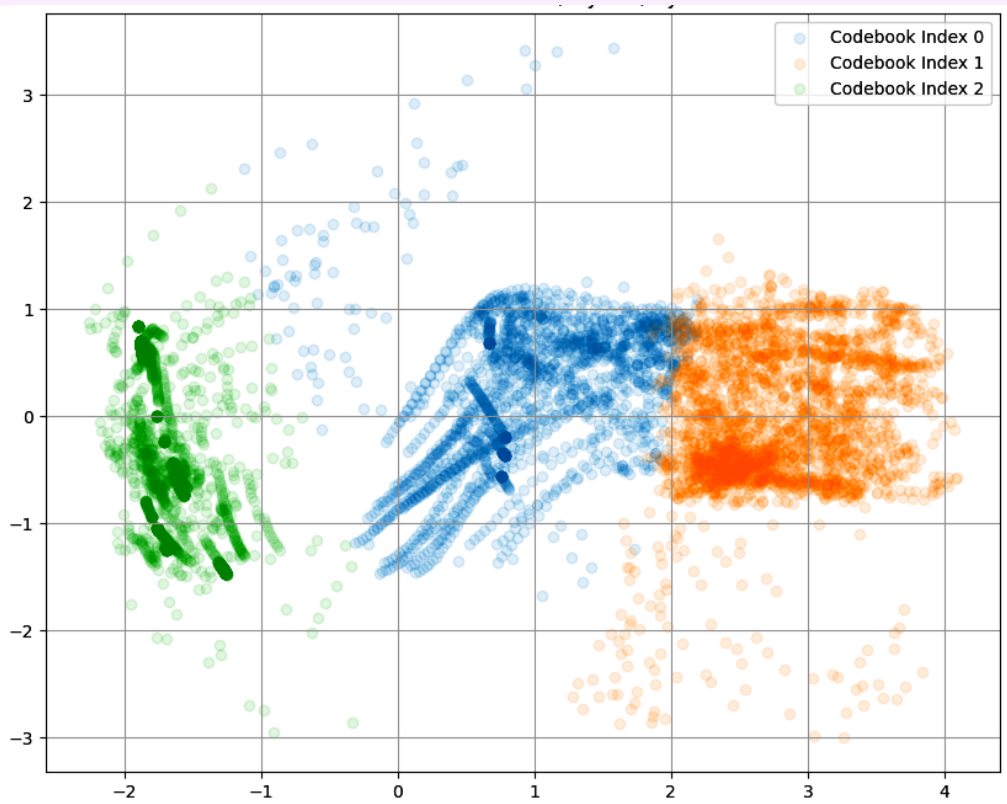

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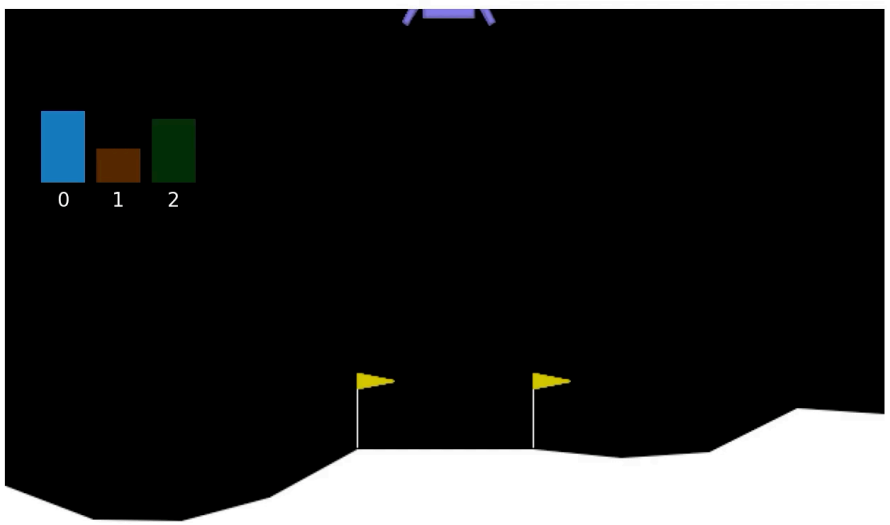

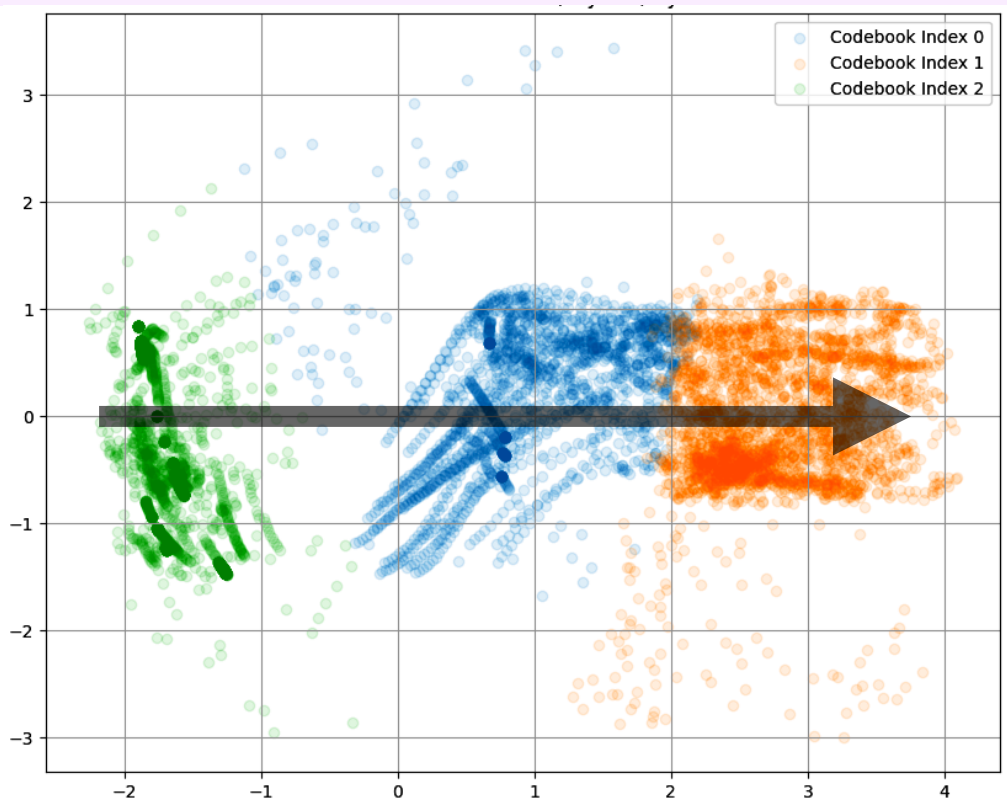

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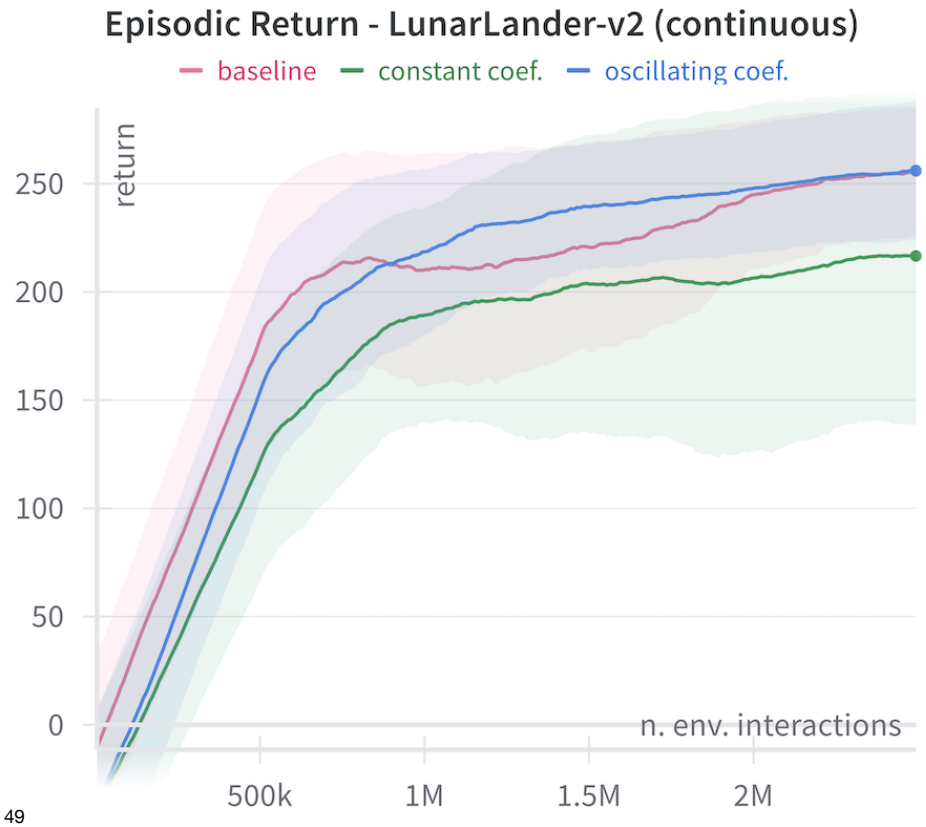

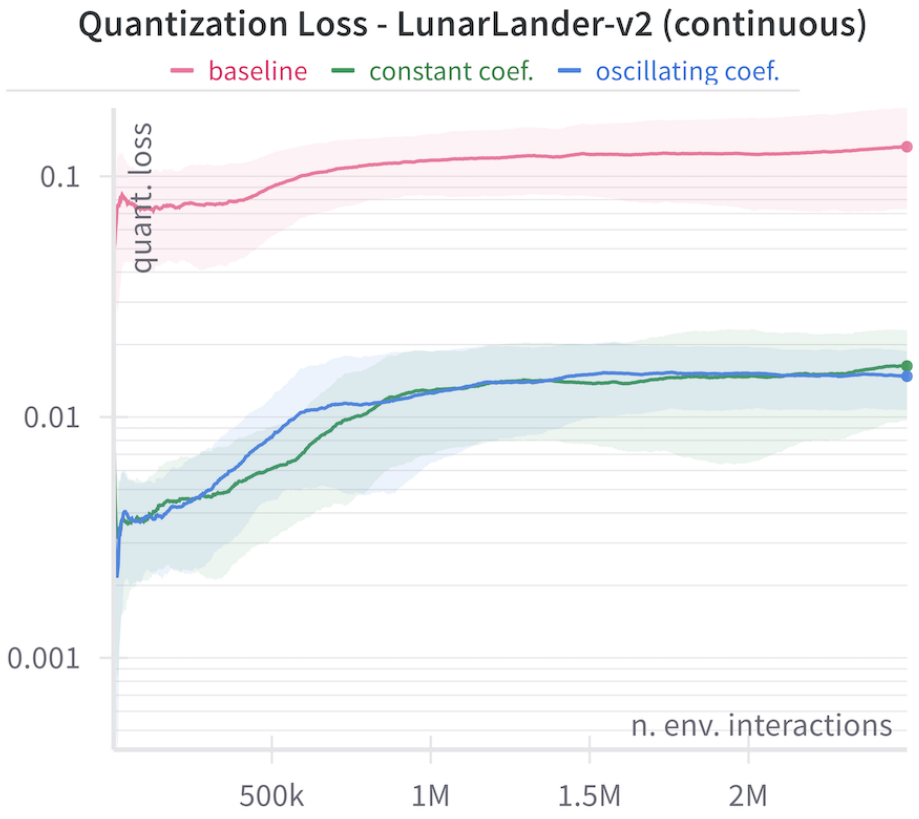

Summary

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Thank you!